

Comprehensive User Guide for Sensitive Data Lockout Tool

1. Introduction to the Tool

The Sensitive Data Locker is a software application designed to provide robust solutions for encrypting and decrypting sensitive data, ensuring that your personal and professional information is protected from unauthorized access. This tool was developed to meet the growing need for cybersecurity in an increasingly data-driven world. The tool aims to enable users to protect their important files and documents easily and effectively, through a simple and straightforward user interface.

The tool relies on strong encryption algorithms to ensure data confidentiality, and offers features such as user management and activity logging, making it suitable for both personal and professional use. The tool is designed to be easy to use, even for users who don't have a deep technical background in crypto, while providing a high level of security.

The tool includes two main tabs: "Data Lock" to encrypt files, and "Data Decryption" to decrypt them. The tool also provides a feature to verify authorized users, which adds an extra layer of security to access encrypted data.

2. Operating Requirements

To run the Sensitive Data Locker tool, you must have the following requirements on your operating system:

Operating system:

- Windows 10 or later (including Windows 11).
- macOS (newer versions).
- Linux (modern distributions such as Ubuntu).

Python Environment:

- Python 3.6 or later. The latest stable version of Python 3 (such as Python 3.10, 3.11, 3.12, or 3.13) is recommended.
- You can check your installed version of Python by opening the Command Prompt or Terminal and typing 'python --version' or 'python3 --version'.

Required libraries (if you are running from source code):

- 'tkinter': Usually pre-installed with Python, but you may need to install it separately .
- 'Cryptography': This library is essential for encryption and decryption operations. They can be installed using 'pip':

```
pip install cryptography
```

- PyInstaller (if you want to build an executable version yourself):

* If you plan to create an executable version of the application, you will need to install PyInstaller:

```
pip install pyinstaller
```

3. Operating Instructions

The Sensitive Data Locker can be run in two main ways: through the executable (recommended) version or through the source code.

3.1. Running the Executable Version

The executable version is a single file that does not require the installation of Python or any additional libraries, making it easy to run on Windows systems.

1. Unzip the file: If you receive the application as a compressed file (e.g. '.zip' or '.7z'), unzip it to a folder of your choice (for example, on your desktop or in a folder dedicated to apps).
2. Launch the application: Inside the folder you unzipped, look for the executable. It is usually named 'DataSecurityApp.exe' or 'main_app.exe' located in the dist folder.
3. Double-click: Double-click the executable to launch the app.

Important: If you encounter the message "This app can't run on your PC", it means that the executable file you are trying to run is not compatible with your operating system. In this case, you must build the executable version yourself on your device (see Section 3.3).

3.2. Run from source code (for developers and advanced users)

If you prefer to run the app from source code, or if you're having problems with the executable version, you can follow these steps:

1. Make sure you have Python and the required libraries installed: (see Section 2: Operating Requirements).
2. Unzip the source files: If you receive the source files as a zip file, unzip it to a folder of your choice.
3. Open the Command Prompt or Terminal:
4. Go to the app folder: Use the 'cd' command to go to the folder that contains the 'main_app.py'.

Example:

```
cd C:\Users\YourUsername\Desktop\DataSecurityApp
```

(Replace 'YourUsername' with your user's actual name, and the path with the correct path of the app folder.)

5. Launch the app: Type the following command and press 'Enter':

```
python main_app.py
```

Or if you have multiple versions of Python:

```
python3 main_app.py
```

3.3. Build the executable version yourself (for advanced users)

If you want to build your own executable to ensure optimal compatibility with your system, follow these steps:

1. Make sure you have Python and PyInstaller installed: (see Section 2: Operating Requirements).
2. Go to the App Folder: Open the Command Prompt and navigate to the folder that contains the 'main_app.py' file.
3. Clean up old build folders (optional but recommended):

Open cmd

```
rmdir /s /q build dist
```

```
del.spec
```

4. Build the application: Use the following command. This command will create a single executable file ('.exe') in the 'dist' folder, and will contain all the necessary files:

Open cmd

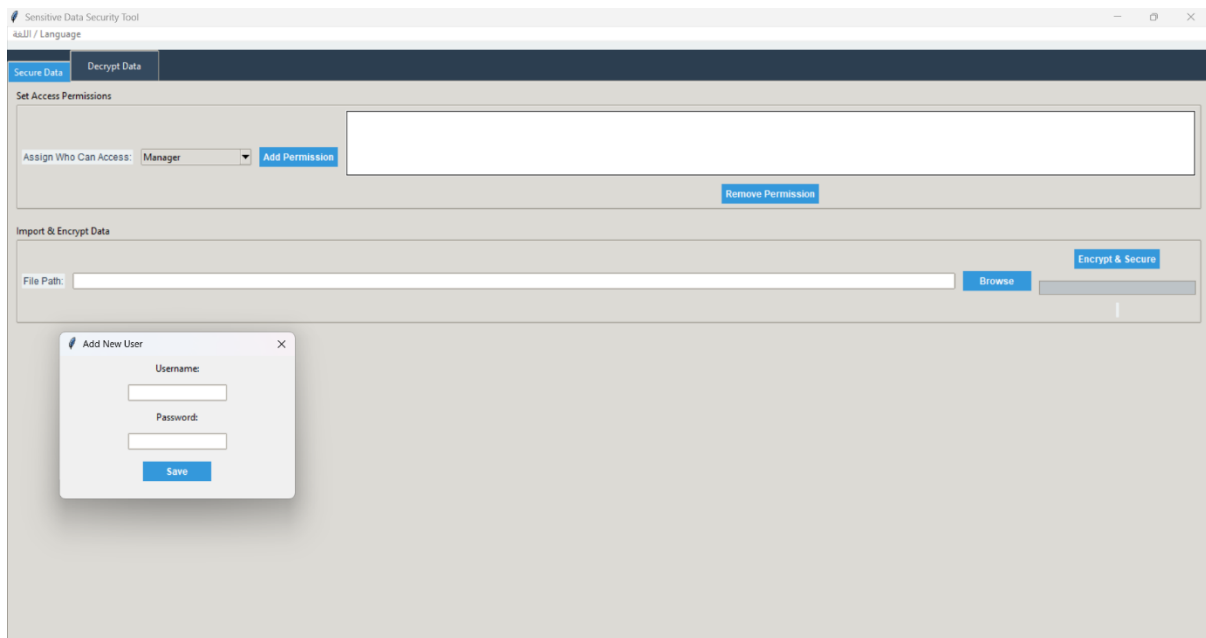
```
pyinstaller --noconfirm --onefile --windowed --name DataSecurityApp --add-data "encryption_module.py;." --add-data "user_management.py;." --add-data "users_db.json;." --add-data "activity_log.txt;." --add-data "icon.ico;." main_app.py
```

5. Run the build: After the build is complete, you will find the executable file 'DataSecurityApp.exe' inside the 'dist' folder (which will be inside the application folder). You can run it by double-clicking on it.

4. User Manual

4.1. Data Security (Encryption)

This tab is dedicated to encrypting files, and it has clear controls to make the process easier. When you open this tab, you'll see the following interface:



Explaining the interface elements and how to deal with them:

1. "Assign Access" section:

This section contains fields for managing user permissions. You can select the type of permission (such as "Manager") and add a new username and password, or remove certain permissions.

Manager: Assign who can access field: Allows you to specify users who have certain permissions.

Add Permissions Button: Used to add a new username and password to the user (Hossam & 123)

Remove Permission Button: Used to remove permission from a user.

Note: In the right corner of the screen, there is a button to switch the language between Arabic and English that can be used as needed.

2. "Add New User" pop-up window:

- This window appears when you need to add a new user to the system. Requires you to enter your username and password.
- Username field: Enter your new username here.
- Password field: Enter the password for the new user here.
- Save button: Used to complete the process of adding a new user.

3. "Encrypt Data" section:

This is the main section of file encryption.

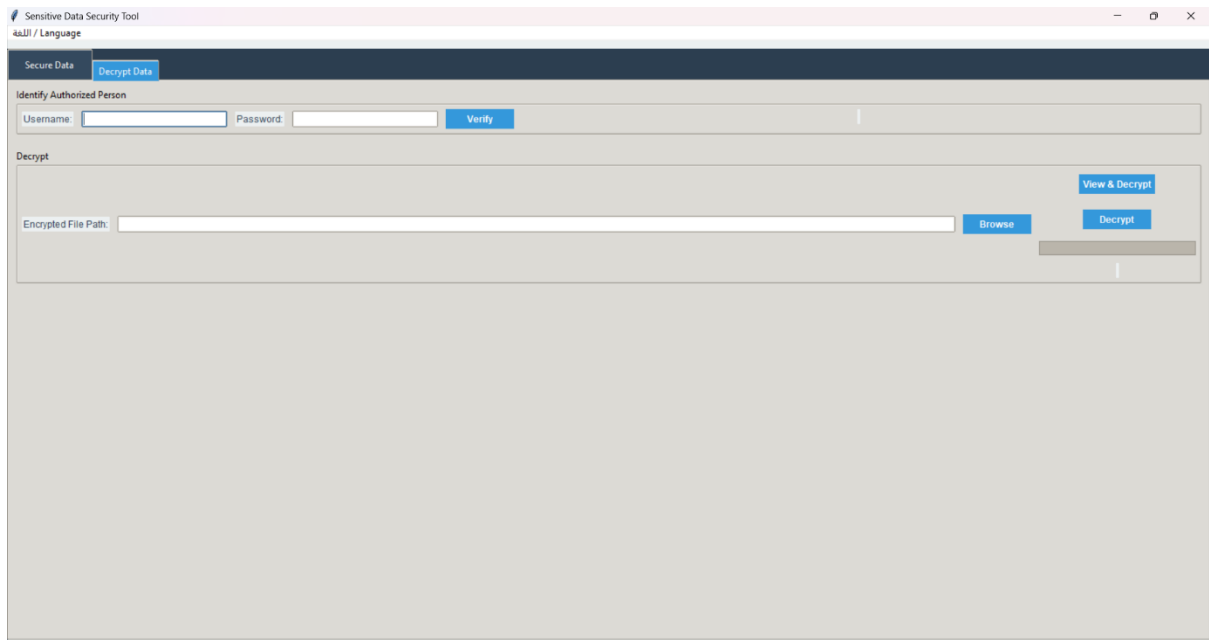
File Path field: Displays the full path of the file chosen for encryption.

Browse button: Click on this button to open the Browse file window and choose the text file you want to encrypt.

Encrypt Button: After selecting the file and entering the encryption key (if applicable, the data icon), click on this button to start the encryption process. The encrypted file will be saved in the same path as the original file with the addition of the '.enc`.

4.2. Data Decryption

This tab is for decrypting files that have been encrypted using the tool. When you open this tab, you'll see the following interface:



Explaining the interface elements and how to deal with them:

1. "Identify Authorized Person" section:

- This section is used to verify the identity of the user authorized to decrypt.
- Username field: Enter your username here.
- Password field: Enter your password here.
- Verify button: Click this button to validate your username and password.
- The tool will encrypt the file and generate a key file ('.key') with the same name as the original file (example: 'document.docx.key'). This key is essential for decryption and must be kept safe.
- The access permissions you selected with the key will be stored in the tool's internal database.

Note: Your permissions must match the permissions specified for the encrypted file.

2. "Decryption" section:

- Click on the 'Browse' button to choose the encrypted file (which usually ends with '.encrypted').
- View and Decrypt Button: This button provides a convenient function to decrypt the file and view its content immediately after decryption. When you click on it, the app will decrypt the file and display its contents in a pop-up window and for decryption without displaying Click the "Decrypt" button: Click on this decrypt button. The decrypted file will be saved in the same path as the encrypted file with the addition of an '.dec'.
- Very important note: Zooming in on the interface ensures that all the details are displayed completely and all the elements are displayed correctly.

- If your permissions are correct, the tool will decrypt the file and save it to the same original folder (with the suffix removed'.encrypted').

5. Additional Information

- User management: The tool includes a user management platform, allowing multiple accounts to be created and their roles defined (such as administrator and regular user). This adds an extra layer of security to control who can access the encryption and decryption functions.
- Activity Log: The tool logs all important operations (e.g. login, encryption, decryption) in the activity log. Administrators can review this log to track the use of the tool and detect any suspicious activities.
- Compatibility: The tool is designed to be compatible with major operating systems (Windows, macOS, Linux) when run from source code. Executable versions are specific to the operating system on which it is built.

7. References

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